



COMPUTER NETWORKS

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Department of Computer Science and Engineering

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Transport Layer

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Transport Layer - Roadmap

3.4 principles of reliable data transfer

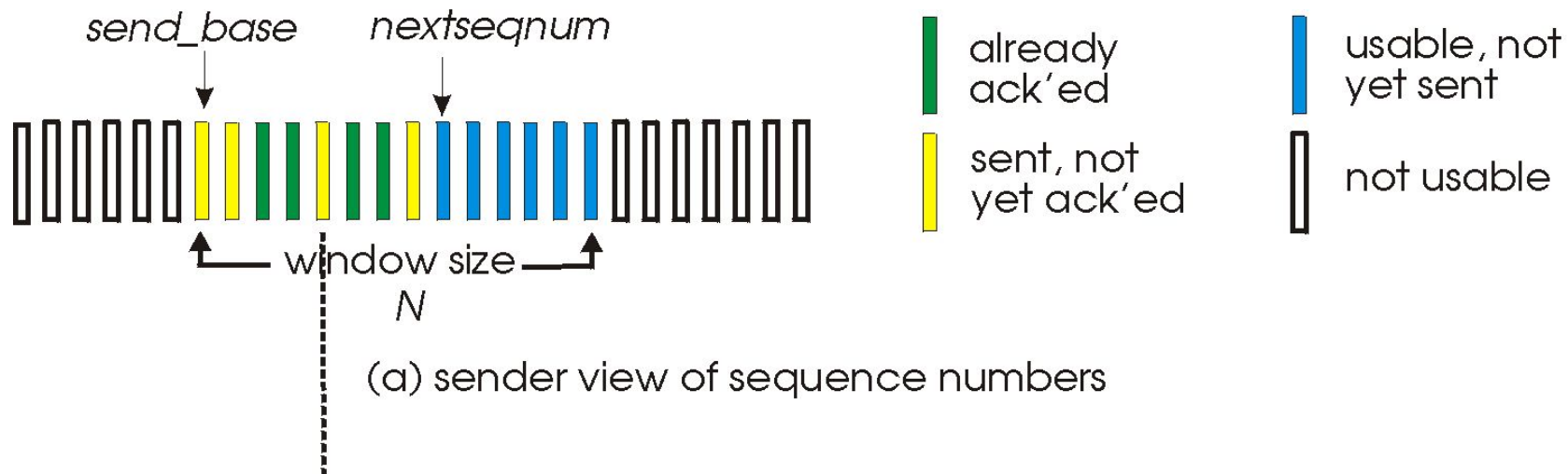
3.5 connection-oriented transport: TCP

- segment structure
- reliable data transfer
- flow control
- connection management

3.6 principles of congestion control

3.7 TCP congestion control

- receiver *individually* acknowledges all correctly received packets
 - buffers packets, as needed, for eventual in-order delivery to upper layer
- sender times-out/retransmits individually for unACKed packets
 - sender maintains timer for each unACKed pkt
- sender window
 - N consecutive seq #s
 - limits seq #s of sent, unACKed packets



Selective Repeat Applet -

<https://computerscience.unicam.it/marcantoni/reti/applet/SelectiveRepeatProtocol/selRepProt.html>

sender

data from above:

- if next available seq # in window, send packet

timeout(n):

- resend packet n , restart timer

ACK(n) in [sendbase, sendbase+N]:

- mark packet n as received
- if n smallest unACKed packet, advance window base to next unACKed seq #

receiver

packet n in [rcvbase, rcvbase+N-1]

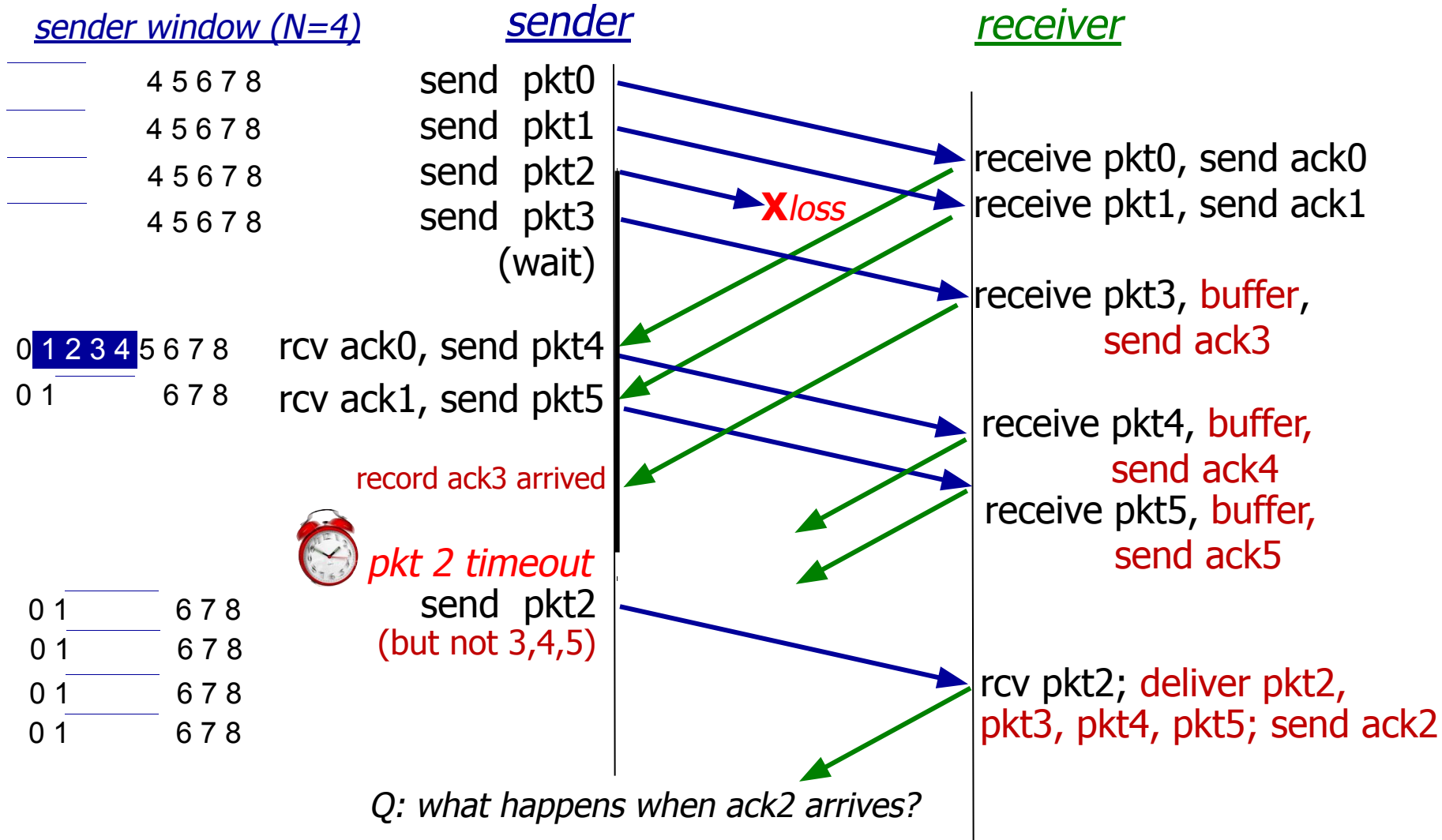
- send ACK(n)
- out-of-order: buffer
- in-order: deliver (also deliver buffered, in-order packets), advance window to next not-yet-received packet

packet n in [rcvbase-N, rcvbase-1]

- ACK(n)

otherwise:

- ignore



In GB3, if every 5th packet that is transmitted is lost and if we have to send 10 packets, then how many transmissions are required?

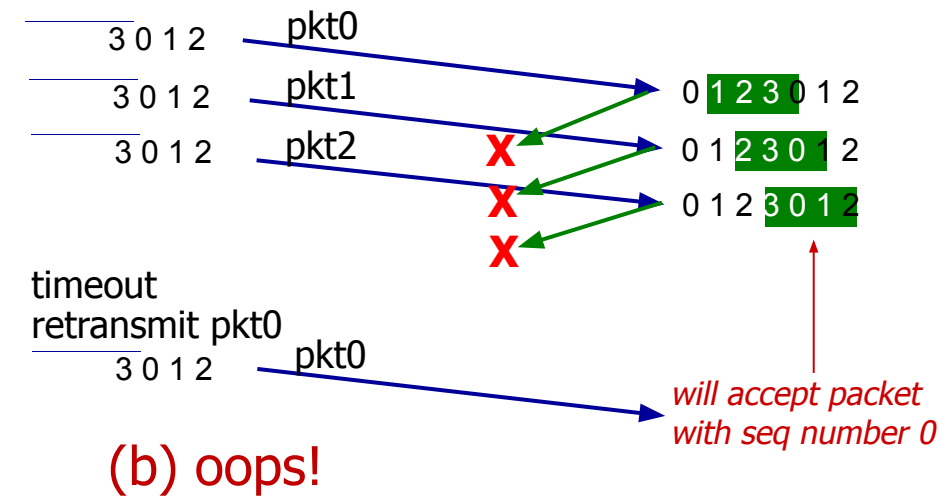
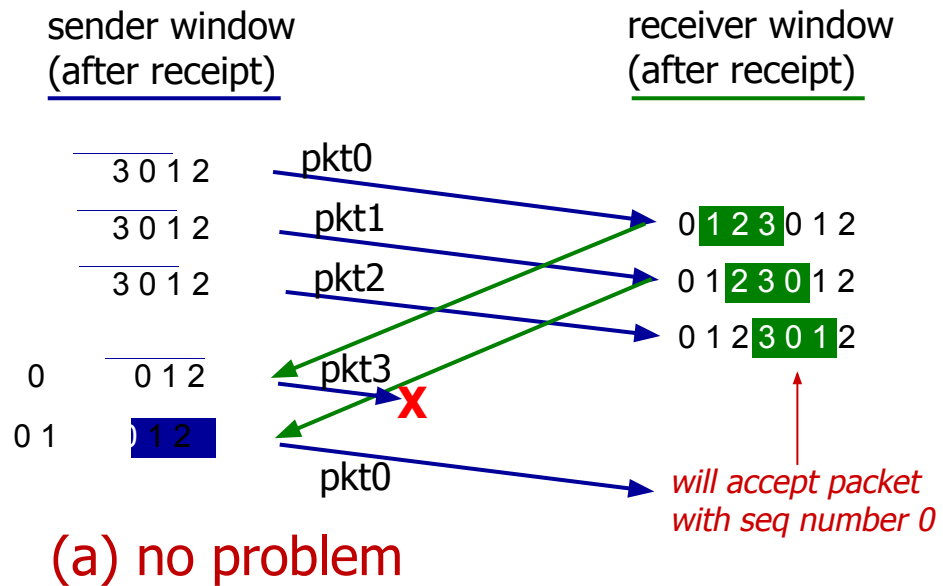
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Selective repeat: a dilemma!

example:

- seq #s: 0, 1, 2, 3 (base 4 counting)
- window size=3



Selective repeat: a dilemma!

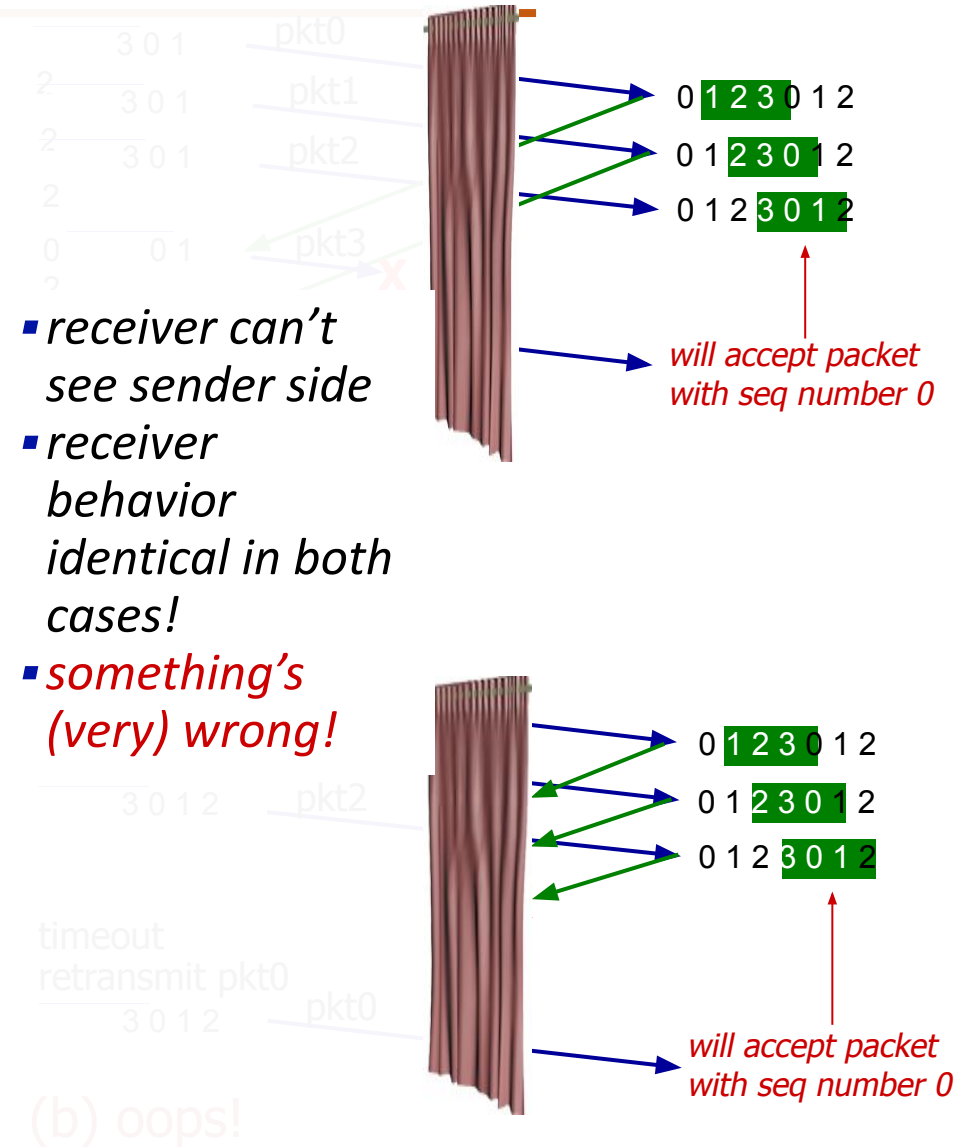
example:

- seq #s: 0, 1, 2, 3 (base 4 counting)
- window size=3

Q: what relationship is needed between sequence # size and window size to avoid problem in scenario (b)?

sender window
(after receipt)

receiver window
(after receipt)





THANK YOU

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